DIAGNOSTIC PERFORMANCE OF ENHANCED DEEP LEARNING ALGORITHMS FOR SKIN CANCER DETECTION USING 3D TOTAL BODY PHOTOGRAPHY (3D-TBP)

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Introduction

- Skin cancer is a growing global health concern: 330,000+ new melanoma cases and 60,000 deaths worldwide in 2022 (IARC, 2022).
- In the U.S., over 9,500 diagnosed daily; more than 2 die every hour (Skin Cancer Foundation, 2023).
- Melanoma accounts for 1% of cases but causes most skin-cancer-related deaths.
- Disparities exist among racial and ethnic groups:
  - Black individuals: 1–2% of cases, often delayed diagnosis, worse prognosis (SERO, 2023).
  - Melanoma often appears in less sun-exposed areas: soles, palms, nail beds (UT Southwestern, 2024).
  - 26% higher mortality in Black men vs. White men; five-year survival: 52% vs. 75% (Time, 2024).
- Causes of disparities:
  - Underrepresentation of darker skin in medical education.
  - Lack of public awareness of non-traditional lesion sites.
- Solution requires: inclusive medical training, outreach, and equitable diagnostic technologies (American Cancer Society, 2023).

Motivation

- Skin cancer is **highly curable when detected early** but often **deadly when diagnosed late**.
- **Millions worldwide lack access** to dermatologists and specialized care.
- Existing **AI tools underperform on darker skin tones**, creating health inequities.
- **My work aims to build fair, accurate, and accessible detection tools for all populations.**

Problem Statement

- Limited access to dermatologic expertise in underserved regions.
- High mortality due to late diagnosis of skin cancer.
- Limitations of traditional AI models: dependency on high-resolution images, lack of skin tone diversity.
- Need for equitable, generalizable AI solutions using real-world 3D-TBP images.

Objective of the Study

General Objective: The main objective of this thesis is to design and implement two advanced deep learning-based classification models for detecting skin cancer lesions using cropped 3D Total Body Photography (3D-TBP) data.

Specific Objectives:

- Gather and preprocess 3D Total Body Photography (3D-TBP) images of skin lesions.
- Implement two deep learning models: EfficientNetV2 (EffV2) and CNN-ANN hybrid.
- Design and compare the performance of both models in classifying skin lesions.
- Address class imbalance using data augmentation and SMOTE.
- Validate model performance across diverse skin tones.

Research Questions

- How to gather and preprocess 3D-TBP skin lesion images effectively?
- How to implement EfficientNetV2 and CNN-ANN models for skin lesion classification?
- How to design and compare the performance of EfficientNetV2 and CNN-ANN models?
- How to address class imbalance using data augmentation and SMOTE?
- How to validate model performance across diverse skin tones?

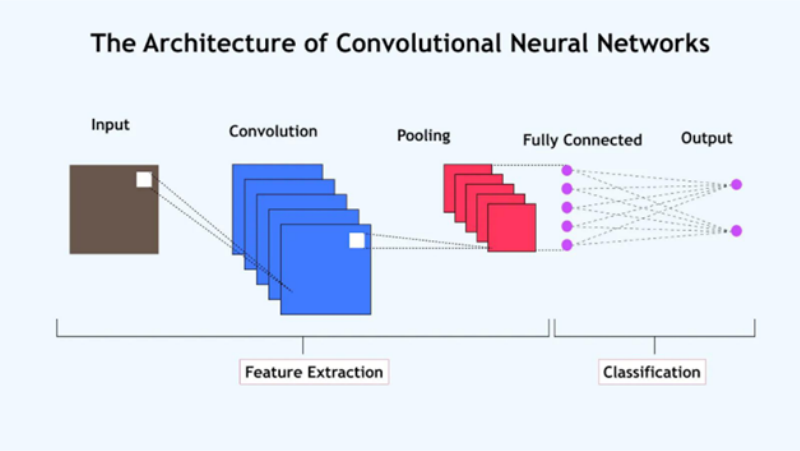
Methodology

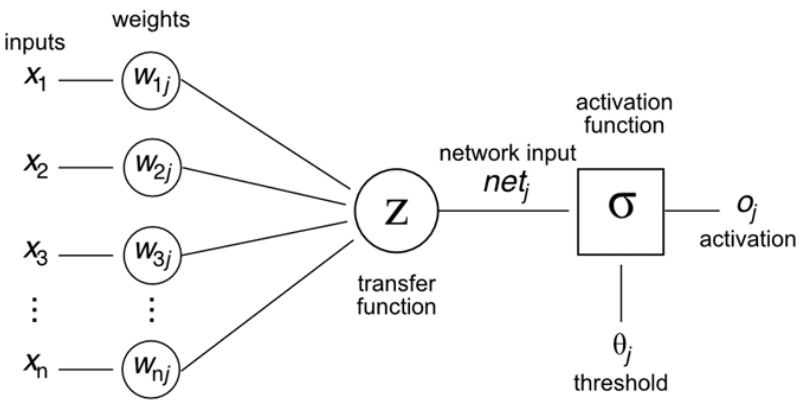
Dataset: ISIC2024 SLICE-3D (400,000+ 3D-TBP lesion crops, metadata included)

Experiments:

- **Model 1:** Multimodal EfficientNetV2 (image + metadata) Trained under two regimes: Fixed LR (0.001, 0.01, 0.1, CosineAnnealing scheduler) and Dynamic LR (CosineAnnealing scheduler)
- **Model 2:** CNN-ANN hybrid (image + metadata inputs) with SMOTE to balance class distribution
- **Training Strategies:** Stratified Group K-Fold (patient-level division), class weighting, data augmentation (flip, rotation, zoom, brightness), and dropout regularization

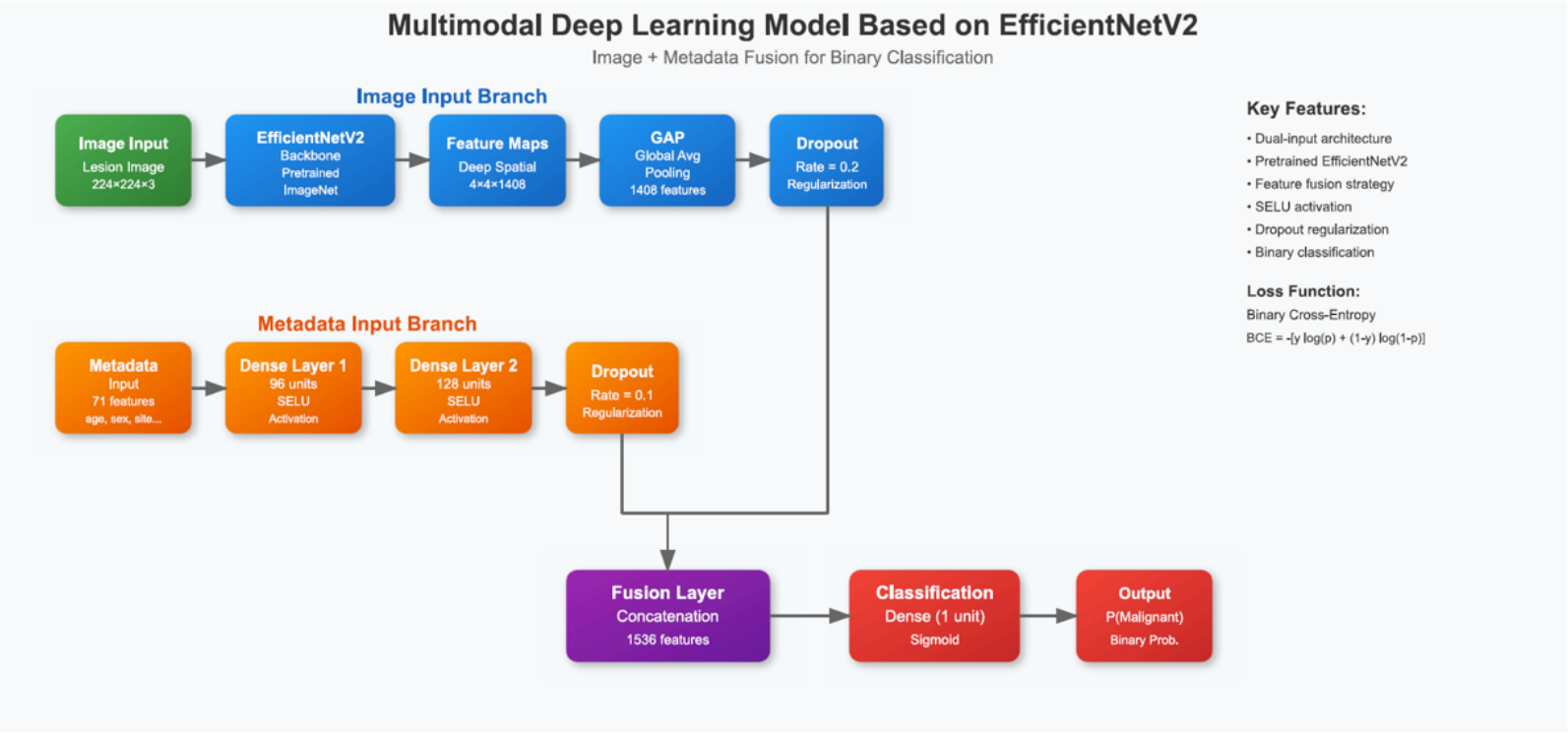
Current Model Architecture





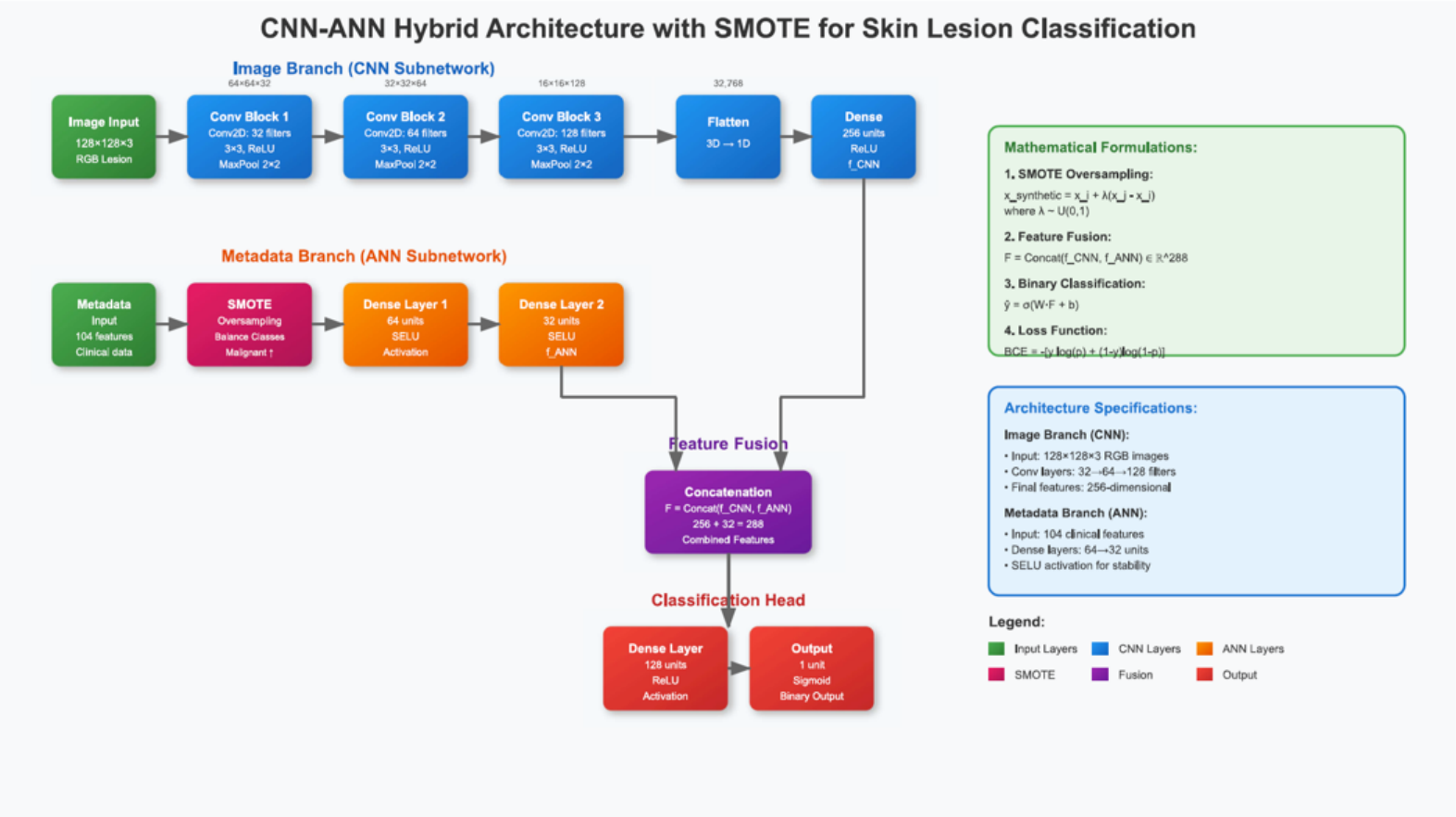
Proposed Model Architectures

Model 1: EfficientNetV2 + Metadata Dual-branch structure combining image and metadata inputs. Includes Global Average Pooling (GAP), dropout, and dense layers for classification.



Proposed Model Architectures

Model 2: CNN-ANN Hybrid CNN branch processes the image stream, while the ANN branch processes metadata. Outputs are concatenated and passed through a classifier for final prediction.



Experiment 1: Learning Rate Impact (EfficientNetV2)

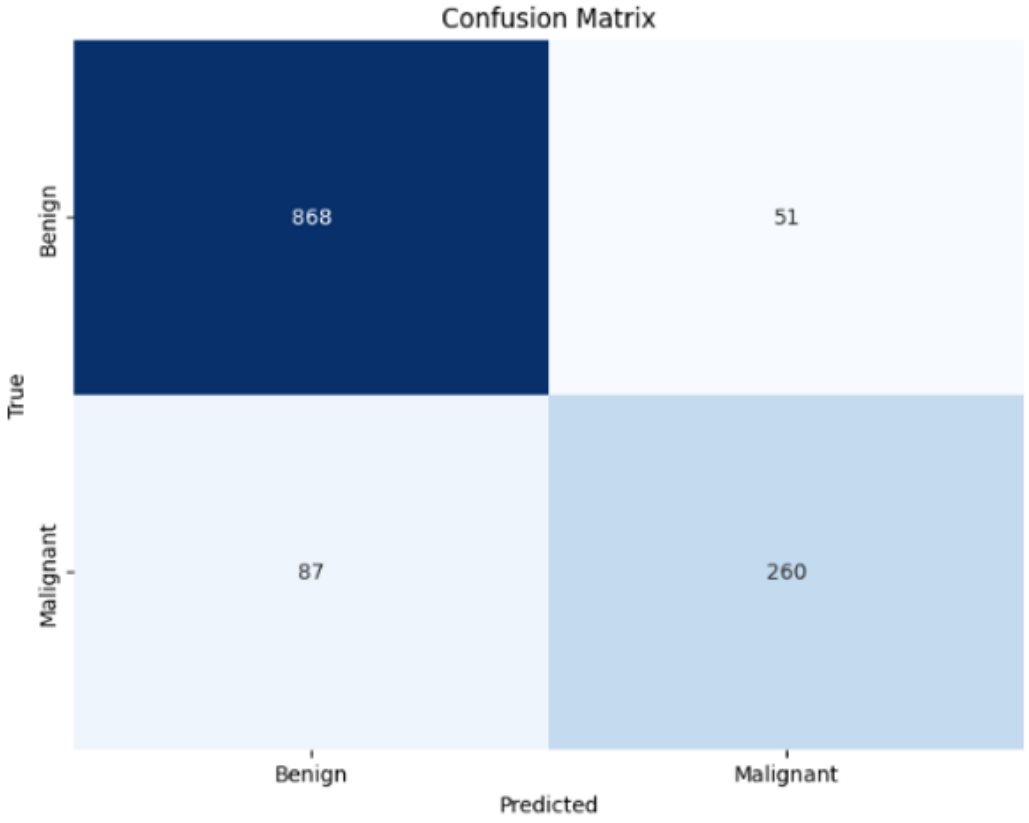
Table 1: Comparative Summary of Learning Rates (EfficientNetV2)

Metric	**LR=0.001**	**LR=0.01**	**LR=0.1**	**Scheduler**
Accuracy	88%	86%	85%	91%
Precision (Benign)	0.90	0.94	0.92	0.91
Precision (Malig.)	0.80	0.71	0.71	0.82
Recall (Benign)	0.93	0.87	0.87	0.92
Recall (Malig.)	0.73	0.84	0.79	0.84
F1 (Malignant)	0.77	0.77	0.75	0.80
AUC	0.946	0.944	0.910	0.957

Experiment 2: Comparative Model Performance

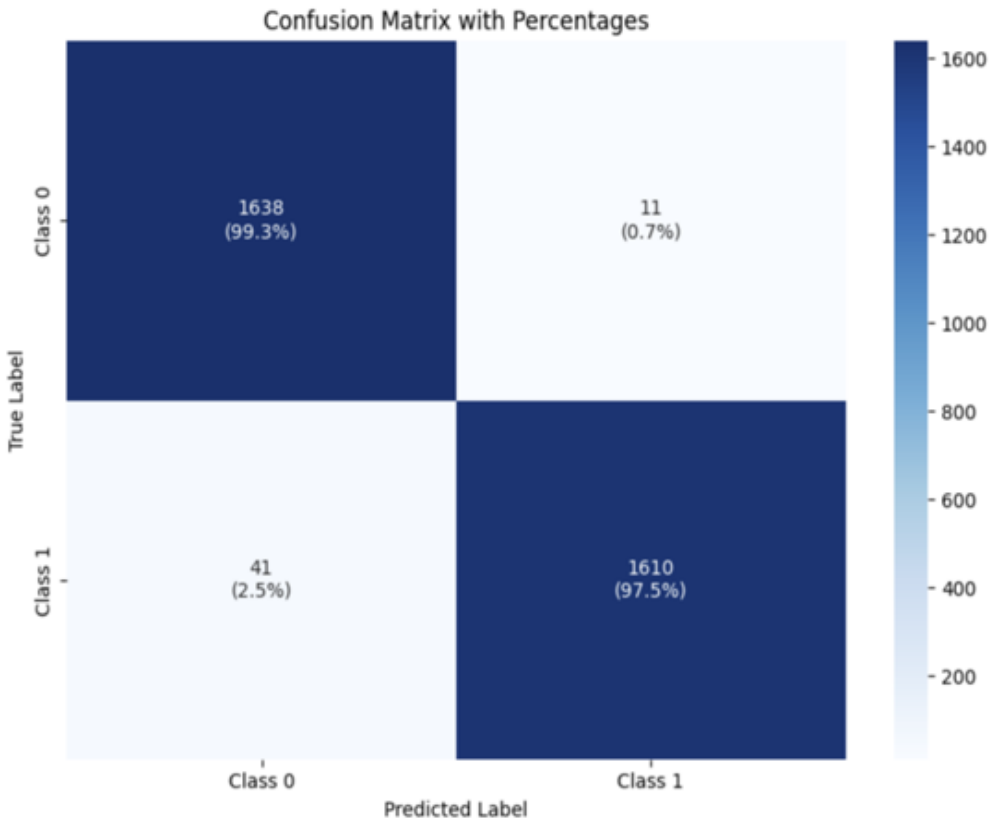
Technique	Dataset	AUC (%)	ACC (%)	SEN (%)	SPE (%)
EfficientNetV2 + Metadata	ISIC 2024 Slice 3D-TBP	95.75	91.00	75.00	93.00
CNN-ANN + SMOTE	ISIC 2024 Slice 3D-TBP	99.62	98.40	99.32	97.52

Confusion Matrix - EfficientNetV2 + Metadata Model



Confusion Matrix of EfficientNetV2 + Metadata Model

Confusion Matrix - CNN-ANN + SMOTE Model



Confusion Matrix of CNN-ANN + SMOTE Model

Performance Evaluation Results

Table 6: Performance evaluation results of state-of-the-art algorithms in skin lesion classification

Technique	**Dataset**	**AUC (%)**	**ACC (%)**	**SEN (%)**	**SPE (%)**	**Reference**
EfficientNet-B7	ISIC 2020	94.8	92.3	91.5	93.2	Balaji & Srinivasan, 2021
ResNet-50 / DenseNet-121	ISIC 2019	91.2	88.6	87.8	89.4	Yildirim, 2020
ResNet-50 / DenseNet-121	PH2	90.5	87.3	85.7	88.9	Sharma & Singh, 2022
Vision Transformer (ViT)	ISIC 2021	95.2	93.6	93.1	92.4	Liu & Sun, 2023
Hybrid CNN-RNN Transformer	Derm7pt	90.2	89.3	93.8	91.1	Zhang & Li, 2024
InceptionV3	HAM10000	92.1	89.8	88.5	90.6	Khan & Goyal, 2020
Deep Ensemble Learning	ISIC 2022	96.3	94.7	93.9	95.5	Patel & Kumar, 2021
Proposed EfficientNetV2 + metadata	ISIC 2024 (Slice 3D-TBP)	95.75	91	75	93	-
Proposed CNN-ANN + SMOTE	ISIC 2024 (Slice 3D-TBP)	99.62	98.42	99.32	97.52	-

Discussion Highlights

- **EfficientNetV2 Strengths:** High generalization to diverse image quality and efficient inference when paired with adaptive learning rates; however, struggled slightly with malignant recall under fixed rates.
- **CNN-ANN + SMOTE Benefits:** Superior sensitivity and AUC (99.6% AUC, 98.4% Acc) achieved in test set, demonstrating that metadata fusion and balanced resampling dramatically improve malignant detection.
- **Metadata Role:** Age, lesion site, and skin-tone information significantly enhanced performance in both models by providing clinical context often overlooked by image-only systems.

Real-World Potential

- **EfficientNetV2 Model:** Ideal as a decision-support tool in clinical centers; efficient, reliable, and excels in high-quality image settings.
- **CNN-ANN + SMOTE Model:** Suitable for semi-autonomous mobile or remote screening; resilient to poor image quality, high recall for malignant cases, and metadata enhances predictions.
- **Impact on Equity:** High-sensitivity performance supports early cancer detection in underserved communities with limited access to dermatologists.

Limitations

- **Dataset Bias:** SLICE-3D, while large, underrepresents very low-quality teledermoscopic images and rare lesion types.
- **Local Data Acquisition:** Rwandan clinical data delayed due to IRB approval, limiting diversity in more darker skin representation.
- **Computational Load:** EfficientNetV2 + metadata and hybrid CNN-ANN models demand significant GPU resources, not optimal for smartphone-level deployment.
- **Autonomy Concern:** EfficientNetV2 model's recall under fixed learning rates may miss malignant lesions if used without expert review.

Future Work

- **Broader Evaluation:** Include ultra-low quality images captured by phones in remote regions to stress-test models.
- **Explainability:** Integrate XAI tools like Grad-CAM or SHAP to highlight lesion regions and boost clinical trust and transparency.
- **Model Light-weighting:** Compress and optimize models (e.g., pruning, quantization) for mobile deployment.
- **Clinical Trial Planning:** Pilot deployment in Rwandan clinics to validate real-world utility and refine with local data.
- **Extended Classification:** Expand from binary detection to multi-class classification of lesion types (e.g., melanoma vs. SCC vs. BCC).

Conclusion

- **EfficientNetV2 with adaptive learning** delivered strong performance with efficient inference.
- **CNN-ANN + SMOTE** demonstrated superior sensitivity and excellent performance even under variable conditions.
- **Dynamic learning rate scheduling** was shown to be a key optimization tool.
- These models represent a significant step toward accessible, AI-driven skin cancer detection, especially in low-resource and telehealth contexts. Future work will focus on expanding dataset diversity, improving explainability, and deploying production-ready mobile systems.

Recommendations

- Adopt adaptive learning rates for better performance.
- Prioritize malignant case detection.
- Expand dataset diversity for robust models.
- Use explainable AI for transparency.
- Validate models in real clinical settings.
- Optimize for mobile and low-resource use.
- Ensure data privacy and regulatory compliance.
- Foster collaboration between clinicians and engineers.
- Move towards multi-class lesion classification.
- Train healthcare workers on AI use.

Thank you

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